

Shenzhen Jimi IoT Co., Ltd.

LL301 Command List

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Change History

Version	Date	Support firmware version	Note
V1.5	4/7/2024	TT73_LL301_WAAP_LA_V4.6_240429.1627 TT73_LL301_WAAP_EU_V4.7.1_240724.1758	1. New release.
V1.6	12/5/2024	TT73_LL301_WAAP_EU_V4.8_241204.1723	1. Change the format. 2. Add network, GPS and Bluetooth sensor related commands.
V1.7	1/6/2025	TT73_LL301_WAAP_EU_V4.8_241210.1858 TT73_LL301_WAAP_EU_V4.7.1_240724.1758	1. Add commands to check device position. 2. Modify command for monitoring.
V1.7.1	4/14/2025	TT73_LL301_WAAP_EU_V4.8_241210.1858	1. Modify the parameter setting range of the "Manual power-off event alert" (EVENTALERT,PWROFF) and "Movement start/stop event alert " (EVENTALERT,MSS) (deleted alarm method 2 "SMS+GPRS+Call").

1 Query-Only Commands

1.1 Check firmware version

VERSION#

Example(s)

Command	Reply
VERSION#	[VERSION]TT73_LL301_WAAP_EU_V4.7_240603.1424

1.2 Check basic parameters of device

PARAM#

Example(s)

Command	Reply
PARAM#	IMEI: 864032050000XXX; SOS: ,; CENTER: ; TimeZone: E,8,0 (AUTO);

1.3 Check comprehensive device information

CHECK#

Example(s)

Command	Reply
CHECK#	IMEI: 864032050088XXX; VERSION: TT73_LL301_WAAP_EU_V4.7_240603.1424; MCU_VER: L341_EF_V24_240130; SERVER: 1,gpsdev.tracksolidpro.com,21102; GET IP: 159.138.88.11; APN: cmnet,;; CSQ[4G]: 20; GPRS: 1; KKS: 1; BAT: 3.76; SOS NUM: ,; IMSI: 460082658600XXX; ICCID: 898604A6102180330XXX; CHARGE: 0; STWAKE: 8; LTEWAKE: 512; MCU: 2024: 12: 3: 2: 14: 0; MODE=2,20,0,1,1,1,1; GNSS_TYPE: ZKW_AT6558D;

1.4 Query device network setting

GPRSSET#

Example(s)

Command	Reply
GPRSSET#	GPRS,ON; Currently use APN: UNINET,,; APN Auto set: ON; SERVER,1,gpsdev.tracksolidpro.com,21102; URL,http: //maps.google.com/maps?q=

1.5 Check current status of device

STATUS#

Example(s)

Command	Reply
STATUS#	Battery: 3.76V,NORMAL; GPRS: Link Up; LTE Signal Level: Strong; GPS: OFF; ACC: OFF; Defense: OFF; Button: remove;

1.6 Check current device location

WHERE#

Example(s)

Command	Reply
WHERE#	Current position! Lat: N29.670080,Lon: E106.491959,Course: 274.70,Speed: 3.13Km/h,DateTime: 2021-08-05 15: 05: 55

1.7 Query Google maps link for current position

URL# / POSITION#

Example(s)

Command	Reply
URL#	<08-05 15: 05>http: //maps.google.com/maps?q=N29.670410,E106.491829
POSITION#	<04-23 10: 50>http: //maps.google.com/maps?q=N29.670288,E106.491585

1.8 Locate now

LJDW#

Request the device to locate immediately.

2 Basic Setting Commands

2.1 Manual APN setting

Set the APN only (name only).

APN, apnname#

Set the APN with authentication credentials (name, user and password).

APN, apnname, user, pwd#

Example(s)

Command	Reply	Remarks
APN#	Currently use APN:,, APN Auto set:ON	You can get the currently set APN and the auto APN enable status in the reply of this command.

2.2 Auto APN setting

ASETAPN, SW#

Parameter(s)

Parameter	Meaning	Range/Format	Default
SW	Enable/disable automatic APN detection	ON / OFF	ON

Example(s)

Command	Reply	Remarks
ASETAPN#	ASETAPN:ON Currently use APN:,,	You can get the auto APN enable status and the current APN in the reply of this command.

2.3 Set server parameters

`SERVER,mode,domainName/IP,port#`

Parameter(s)

Parameter	Meaning	Range/Format	Default
mode	Server address format	0 / 1 mode = 1: Set with domain name mode = 0: Set with IP address	
domainName/IP	Server address		
port	Server port		

Example(s)

Command	Reply	Remarks
SERVER,1,gpsdev.tracksolidpro.com,21100#	SERVER set OK!	Setting server with domain name
SERVER,0,47.243.51.117,21100#	SERVER set OK!	Setting server with IP address
SERVER#	SERVER,1,gpsdev-us.tracksolidpro.com,21100	Check current server parameters.

2.4 Set GMT parameters

`GMT,A,B,C#`

Parameter(s)

Parameter	Meaning	Range/Format	Default
A	East of west of GMT	E / W	E
B	Time zone hour offset	0-12	8
C	Half time zone	0/15/30/45	0

Query command

Command	Function
GMT#	Check current time zone parameters

2.5 Auto time zone setting (work for SMS time zone queries)

ASETGMT, SW#

Parameter(s)

Parameter	Meaning	Range/Format	Default
SW	Enable/disable auto time zone setting	ON / OFF	ON

Query command

Command	Function
ASETGMT#	Query current settings

2.6 Set the heartbeat packet type

HBT, TYPE, A#

Parameter(s)

Parameter	Meaning	Range/Format	Default
A	Heartbeat packet type	0 / 1 0: 0x23 heartbeat packet 1: 0x36 heartbeat packet Note: 0x36 packet includes battery percentage.	0

Query command

Command	Function
HBT,TYPE#	Query current settings

2.7 Set the heartbeat sending interval

HBT, T#

Parameter(s)

Parameter	Meaning	Range/Format	Default
T	Heartbeat packet upload interval	1–10 (minutes)	3

Query command

Command	Function
HBT#	Check the current heartbeat sending interval.

2.8 Set network mode`NETSET,A#`**Parameter(s)**

Parameter	Meaning	Range/Format	Default
A	Network selection mode	0/1/2 0: 2G only 1: 4G only 2: 2G+4G (automatic)	2

Query command

Command	Function	Remarks
NETSET#	Query current network setting	Reply format: NETSET,A,B,C A=0/1/2; Configured network mode B=2G/4G; Current registered network C=Current network operator

2.9 Recording server setting`SERVER4,mode,domainName/IP,port#`**Parameter(s)**

Parameter	Meaning	Range/Format	Default
mode	Server address format	0 / 1 mode = 1: Set with domain name mode = 0: Set with IP address	
domainName/IP	Server address		
port	Server port		

Example(s)

Command	Reply	Remarks
SERVER4,1,gpsdev.tracksolidpro.com,20010#	SERVER4 set OK!	Setting server with domain name
SERVER4,0,47.89.58.215,20010#	SERVER4 set OK!	Setting server with IP address
SERVER4#	SERVER4,1,gpsdev.tracksolidpro.com,20010	Check current recording server parameters.

2.10 SOS number setting**Add one or more SOS phone numbers**

SOS,A,phone number 1,phone number 2,phone number 3#

Delete the phone number based on sequence

SOS,D,sequence number 1, sequence number 2,sequence number 3#

Delete the matching SOS phone number

SOS,D,phone number#

Check the current SOS number settings

SOS#

2.11 Set monitoring permission

MONPERMIT,X#

Parameter(s)

Parameter	Meaning	Range/Format	Default
X	Phone call permission	0 / 2 / 3 0: Disabled (no incoming calls will activate voice monitoring) 2: Only calls from pre-configured SOS numbers activate voice monitoring 3: Calls from any number activate voice monitoring	2

2.12 Recording duration setting (only work with mode1/mode2)

LY, T#

Upon receiving the command, the device will take an audio recording of the specified time length and uploads the audio data to the recording server (server4) in Packet 0x8A.

Parameter(s)

Parameter	Meaning	Range/Format	Default
T	Recording duration	10–300 (seconds)	60

Query command

Command	Function	Remarks
LY#	Query current network setting	

2.13 Remote listen-in activation (only work with mode1/mode2)

JT#

When this command is received via SMS from a pre-configured SOS number, the device will automatically initiate a voice call back to that SOS number. Answering this call allows the SOS number user to listen to the device's surroundings (voice monitoring)

2.14 Continuous audio recording setting

CXLY, A, B#

When enabled, the device repeatedly records audio segments of the given duration (B), uploads the completed segment, and then immediately starts recording the next segment. The recording continues until a disable command is received.

Parameter(s)

Parameter	Meaning	Range/Format	Default
A	Enable/disable continuous recording	ON / OFF	OFF
B	Recording segment duration	10–120 (seconds)	30

Dependency/Notes

- (1) Recording does not occur during the upload process.
- (2) The continuous recording loop can only be stopped by one of the following events:
 - ① The specific disable command is received: `CXLY, OFF#`
 - ② The device undergoes a full shutdown.
 - ③ The device enters a "fake shutdown" mode.

3 Report Setting Commands

3.1 Set GNSS mode

GPSPMODE, A#

Parameter(s)

Parameter	Meaning	Range/Format	Default
A	GNSS mode	A= 0/1/2 0: BDS only 1: GPS only 2: GPS + BDS	2

3.2 Set AGPS status

AGPS, SW#

Parameter(s)

Parameter	Meaning	Range/Format	Default
SW	Enable/disable AGPS Note: This may increase mobile data consumption	ON / OFF	ON

3.3 Work mode setting

Timing mode

MODE, 1, T, A, B, C#

Parameter(s)

Parameter	Meaning	Range/Format	Default
1	Working mode	1: Timing mode	
T	Location data upload interval	0, 10–86400 seconds 0: Disable upload	No default value (must be set)
A	GPS switch	0=GPS off 1=GPS on	1
B	WIFI switch	0=WIFI off	1

Parameter	Meaning	Range/Format	Default
		1=WIFI on	
C	LBS switch	0=LBS off 1=LBS on	1

Intelligent mode

`MODE, 2, T1, T2, A, B, C, D#`

Parameter(s)

Parameter	Meaning	Range/Format	Default
2	Working mode	2: Intelligent mode	
T1	Location data upload interval in moving state	10–86400 seconds	20
T2	Location data upload interval in static state	0,180–86400 seconds 0: Disable upload when static (available when GPSDUP is enabled)	0
A	GPS switch	0=GPS off 1=GPS on A = 1 (fixed in this mode). If A = 0, it is reserved for future use.	1
B	WIFI switch	0=WIFI off 1=WIFI on	1
C	LBS switch	0=LBS off 1=LBS on	1
D	GPRS switch	0= GPRS off when idle 1=GPRS always on	1

Power saving mode

`MODE, 3, T1, T2, A, B, C#`

Parameter(s)

Parameter	Meaning	Range/Format	Default
3	Working mode	3: Power saving mode	
T1	Initial wakeup time	Format: HH: MM	
T2	Wakeup interval	5–43200 (minutes)	No default value (must be set)
A	GPS switch	0=GPS off	1

Parameter	Meaning	Range/Format	Default
		1=GPS on	
B	WIFI switch	0=WIFI off 1=WIFI on	1
C	LBS switch	0=LBS off 1=LBS on	1

Query current working mode of device

MODE#

3.4 Upload last known valid position when the device becomes static or cannot get a new fix

GPSDUP, SW#

Note: Only valid in MODE2.

Parameter(s)

Parameter	Meaning	Range/Format	Default
SW	Enable/disable uploading the last known position when static	ON / OFF	OFF

Query command

Command	Function
GPSDUP#	Query current settings

4 Security Setting Commands

4.1 Command password setting

A combination of numbers and letters are supported. The password should be at least 1 character and no more than 19 characters. Default: 666666;

Enable command password check

PWDSW, ON#

Disable command password check

PWDSW, password, OFF#

Note: If you forget the password, please contact Jimi IoT.

Change command password

PASSWORD, <P1>, <P2>#

Parameter(s)

Parameter	Meaning	Range/Format	Default
P1	Current command password		
P2	New command password		

4.2 Manual power off restriction

PWRLIMIT, SW#

Parameter(s)

Parameter	Meaning	Range/Format	Default
SW	Disable (ON) or enable (OFF) manual power off via the button	ON/OFF	OFF

Query command

Command	Function
PWRLIMIT#	Query current setting

4.3 Automatic power-on on charging

[AUTOSTART, SW#](#)

Parameter(s)

Parameter	Meaning	Range/Format	Default
SW	Enable/disable automatic power-on when charging begins.	ON/OFF	ON

Query command

Command	Function
AUTOSTART#	Query the automatic power-on status

4.4 Remote shutdown

[SHUTDOWN#](#)

The device will power off immediately after receiving the command if [PWRLIMIT](#) is set to OFF.

4.5 Factory reset

[FACTORY#](#)

Restore all settings to default except APN/SERVER/ACC settings.

4.6 Reboot device

[RESET#](#)

The device will reboot in 20s after receiving the command.

4.7 Query server address lock status

[QLOCKIP#](#)

Query the status of the server address lock feature. Contact Jimi IoT if you need to enable the server lock.

5 Alarm Setting Commands

5.1 Set dialing attempts

CALL, N#

Parameter(s)

Parameter	Meaning	Range/Format	Default
N	Times the device will attempt to cycle through dialing all configured SOS numbers when an alarm triggers a call action	1-3	3

Query command

Command	Function
CALL#	Query current setting

5.2 Manual power-off event alert

EVENTALERT, PWROFF, SW, B#

Parameter(s)

Parameter	Meaning	Range/Format	Default
SW	Enable/disable alert when powered off via button.	ON/OFF	ON
B	Alarm method	0/1 0= GPRS only 1=SMS+GPRS	0

Query command

Command	Function
EVENTALERT,PWROFF#	Query current setting

5.3 Movement start/stop event alert (Only work with MODE2)

EVENTALERT, MSS, SW, B#

Parameter(s)

Parameter	Meaning	Range/Format	Default
SW	Enable/disable movement start/stop event alert.	ON/OFF	OFF
B	Alarm method	0/1 0= GPRS only 1=SMS+GPRS	0

Query command

Command	Function
EVENTALERT,MSS#	Query current setting

5.4 Upper cover open alarm (Tamper Event 2)

REMALMF, SW, M#

Parameter(s)

Parameter	Meaning	Range/Format	Default
SW	Enable/disable upper cover open alarm (when enabled and triggered, the device enters MODE2 for 20 minutes)	ON/OFF	ON
M	Alarm method	0/1 0= GPRS only 1=SMS+GPRS	0

Query command

Command	Function
REMALMF#	Query current setting

5.5 Removal alarm

REMALM, SW, M#

Parameter(s)

Parameter	Meaning	Range/Format	Default
SW	Enable/disable device removal alarm (when enabled and triggered, the device enters MODE2 for 20 minutes)	ON/OFF	ON
M	Alarm method	0/1/2/3 0= GPRS only 1 = SMS+GPRS 2 = GPRS+SMS+Call 3 = GPRS+Call	0

Query command

Command	Function
REMALM#	Query current setting

5.6 Cancel tamper alarm mode

CANCEL#

The device will exit the temporary tracking mode (MODE2) entered due to a tamper alarm (REMALMF or REMALM) after receiving the command. The device reverts to its previously configured work mode.

5.7 Set sensitivity of sensor

LEVEL, A#

Parameter(s)

Parameter	Meaning	Range/Format	Default
A	Sensitivity range The greater the value, the less the sensitivity.	1-5	3

Query command

Command	Function
LEVEL#	Check the current sensitivity level of the motion sensor.

5.8 Vibration alarm (only work with MODE2)

SENALM, SW, M#

Parameter(s)

Parameter	Meaning	Range/Format	Default
SW	Enable/disable vibration alarm	ON/OFF	OFF
M	Alarm method	0/1/2/3 0 = GPRS only 1 = SMS+GPRS 2 = GPRS+SMS+Call 3 = GPRS+Call	0

Query command

Command	Function
SENALM#	Check current setting.

5.9 Low battery alarm

BATALM, SW, M#

Parameter(s)

Parameter	Meaning	Range/Format	Default
SW	Enable/disable low battery alarm (triggers below 3.55V)	ON/OFF	ON
M	Alarm method	0/1 0= GPRS only 1=SMS+GPRS	0

Related commands

Command	Function
BATALM,OFF#	Disable low battery alarm
BATALM#	Check current setting.

6 Sensor Setting Commands

6.1 Bluetooth scanning control

`BTSCAN, SW#`

Parameter(s)

Parameter	Meaning	Range/Format	Default
SW	Enable/disable Bluetooth scanning	ON/OFF	OFF

6.2 Bluetooth sensor MAC address binding

`BTMAC, ADD, MAC1, MAC2, MAC3, .....MAC10#`

The device can bind up to 10 Bluetooth sensor MAC addresses. The device will only interact with bound sensors.

Note: When in MODE1/MODE2, scanning is always active; while in MODE2, scanning is only active when the device wakes up.

Parameter(s)

Parameter	Meaning	Range/Format	Default
ADD	Action to add MAC address(es)		
MAC1...MAC10	MAC addresses to add		

Related commands

Command	Function
BTMAC, DELALL#	Delete all bound MAC addresses.
BTMAC#	Query currently-bound MAC addresses.

6.3 Temperature/humidity data reporting interval

TEMPTIMER, T#

The device enables Bluetooth scanning and reports data received from bound Bluetooth temperature and humidity sensors to the server at T intervals.

Parameter(s)

Parameter	Meaning	Range/Format	Default
T	Data upload interval	0, 1–1440 minutes 0: Disable upload	5

Query command

Command	Function
TEMPTIMER#	Query current setting.

6.4 Query current temperature and humidity

TEMP#

The device replies the latest temperature and humidity data read from the bound sensor.

6.5 Temperature/humidity alarm setting

TEMPALM, SW, L1, H1, L2, H2, M, T#

Parameter(s)

Parameter	Meaning	Range/Format	Default
SW	Enable/disable temperature/humidity alarms	ON/OFF	OFF
L1	Low temperature threshold. Alarm triggers if the temperature falls below this value.	255, –30 to 100 (°C) L1=255: Disable low temperature alarm.	255
H1	High temperature threshold. Alarm triggers if the temperature rises above this value.	255, –30 to 100 (°C) H1=255: Disable high temperature alarm.	255
L2	Low humidity threshold. Alarm triggers if the humidity falls below this value.	255, 0 to 100 (%) L2=255: Disable low humidity alarm.	255

Parameter	Meaning	Range/Format	Default
H2	High humidity threshold. Alarm triggers if the humidity rises above this value	255, 0 to 100 (%) H2=255: Disable low humidity alarm.	255
M	Alarm method	0/1 0=GPRS only 1=SMS+GPRS	0
T	Alarm cooldown interval	1–60 (minutes)	2

Query command

Command	Function
TEMPALM#	Query current setting.

6.6 Light sensor alarm setting

LIGHTALM, SW, M, T#

Parameter(s)

Parameter	Meaning	Range/Format	Default
SW	Enable/disable light sensor alarm	ON/OFF	OFF
M	Alarm method	0/1 0=GPRS only 1=SMS+GPRS	0
T	Alarm cooldown interval	0, 1–60 (minutes) 0: No cooldown	0

Query command

Command	Function
LIGHTALM#	Query current setting.

6.7 Door sensor alarm setting

DOORALM, SW, M, T#

Parameter(s)

Parameter	Meaning	Range/Format	Default
SW	Enable/disable door status alarm	ON/OFF	OFF
M	Alarm method	0/1 0=GPRS only 1=SMS+GPRS	0
T	Alarm cooldown interval	0, 1–60 (minutes) 0: No cooldown	0

Query command

Command	Function
DOORALM#	Query current setting.